Corporate Finance

Lecture 8 extras: How a project is evaluated

Camilo Saldarriaga

Finance Department, ESCP Business School, 2026/2027

How a project is evaluated

- A project costs I today and pays a cash flow c at the end of each of the next N years. Its **cost of capital** r is the return investors could earn elsewhere at the same risk (Lecture 6).
- **Way 1: compute a return, then compare it with the cost of capital.** Ask what return the project itself earns: the rate that makes the NPV zero. That rate is the **internal rate of return** (IRR) :

$$\text{NPV} = -I + \frac{c}{1+\text{IRR}} + \frac{c}{(1+\text{IRR})^2} + \cdots + \frac{c}{(1+\text{IRR})^N} = 0$$

$$I = \frac{c}{1+\text{IRR}} + \frac{c}{(1+\text{IRR})^2} + \cdots + \frac{c}{(1+\text{IRR})^N}$$

- Solve for IRR. Accept the project when the IRR is above the cost of capital: the project earns more than the money costs. Here the test is a rate against a rate.

How a project is evaluated (con't)

- **Way 2: discount, then compare amounts.** Do not compute the project's own return. Instead, discount every c at the cost of capital r and add up:

$$\text{PV} = \frac{c}{1+r} + \frac{c}{(1+r)^2} + \cdots + \frac{c}{(1+r)^N} \qquad \text{NPV} = \text{PV} - I$$

- Accept when the NPV is positive: the project is worth more than it costs.
- Here the **discount rate** is the cost of capital. Once it has been used, there is nothing left to compare it with: the test is an amount against an amount, PV against I.

Two equivalent approaches: a return above the cost of capital is exactly what makes the discounted cash flows exceed the investment. The two analyses give the same answer.

What is a leveraged ETF?

- An **ETF** (exchange-traded fund) is a fund you buy and sell like a share. A plain S&P 500 ETF such as **SPY** holds the 500 stocks of the index: put in 100, and you own 100 of stocks. Leverage: none.
- A **leveraged ETF** promises a **multiple of the index's daily return**: **SSO** targets 2 times the daily return of the S&P 500, **UPRO** 3 times.
- How: the fund borrows, or uses futures and swaps that do the borrowing for it. For every 100 you put into SSO, it holds 200 of exposure: your 100 plus 100 borrowed. The borrowing cost and a fee of about 0.9% a year (against 0.09% for SPY) are taken out of the fund.

This is **homemade leverage sold as a product**: the fund borrows for you, exactly as the investor of Lecture 7 who borrowed 500 and bought the unlevered firm. Twice the exposure means twice the risk, so the expected return and the swings both grow.

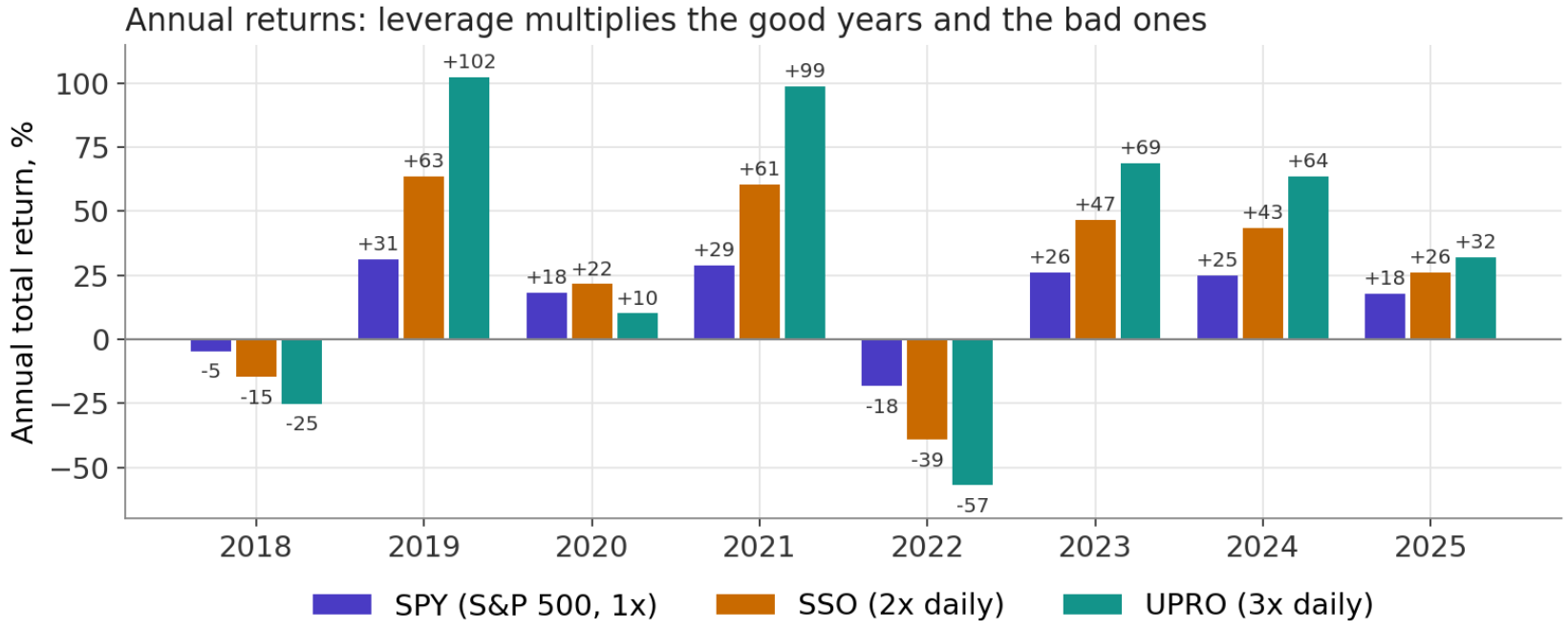
The daily reset: why 2x a day is not 2x a year

- The multiple holds for **one day**. Every evening the fund rebalances so that tomorrow it is again at 2 times. Over longer periods the result drifts.
- Example: the index rises 10% then falls 10%.

Day	Index	2x ETF
Start	100	100
Day 1	$100 \times 1.10 = 110$	$100 \times 1.20 = 120$
Day 2	$110 \times 0.90 = 99$	$120 \times 0.80 = 96$
Result	-1%	-4%

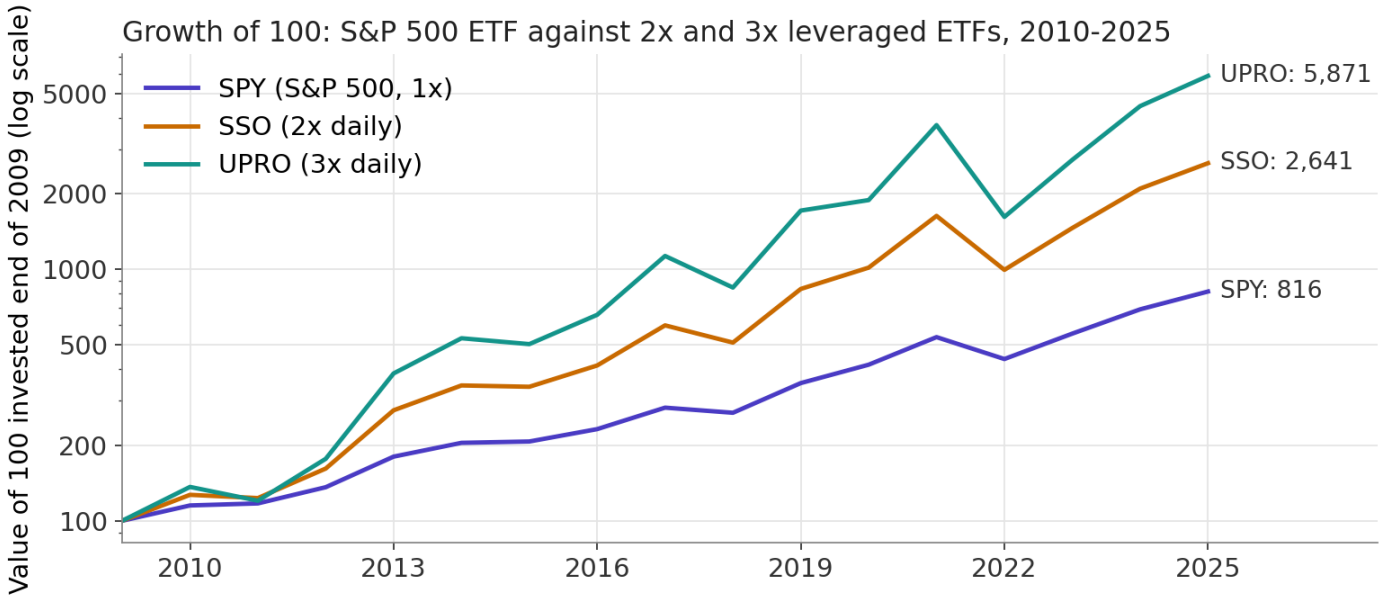
- The fund lost four times the index, not two. Ups and downs eat into a leveraged fund: **volatility decay**. In a steady rise it compounds in the fund's favour; in a choppy market it loses even if the index ends flat.

Recent returns: SPY, SSO and UPRO



Source: annual total returns, dividends reinvested, calendar years, from totalrealreturns.com/n/SPY,SSO,UPRO, read 16 October 2026. Chart by the author.

Recent returns (con't): the long run



- Over a long bull market the leveraged funds won by a wide margin. The price: in March 2020 SPY fell 34% from its peak, SSO 59% and UPRO 77%.
- More leverage, more expected return **and** more risk. The fund creates no value; it repackages risk.

Source: growth of 100 compounded from the annual total returns, and maximum drawdowns, from totalrealreturns.com/n/SPY,SSO,UPRO, read 16 October 2026. Chart by the author.